

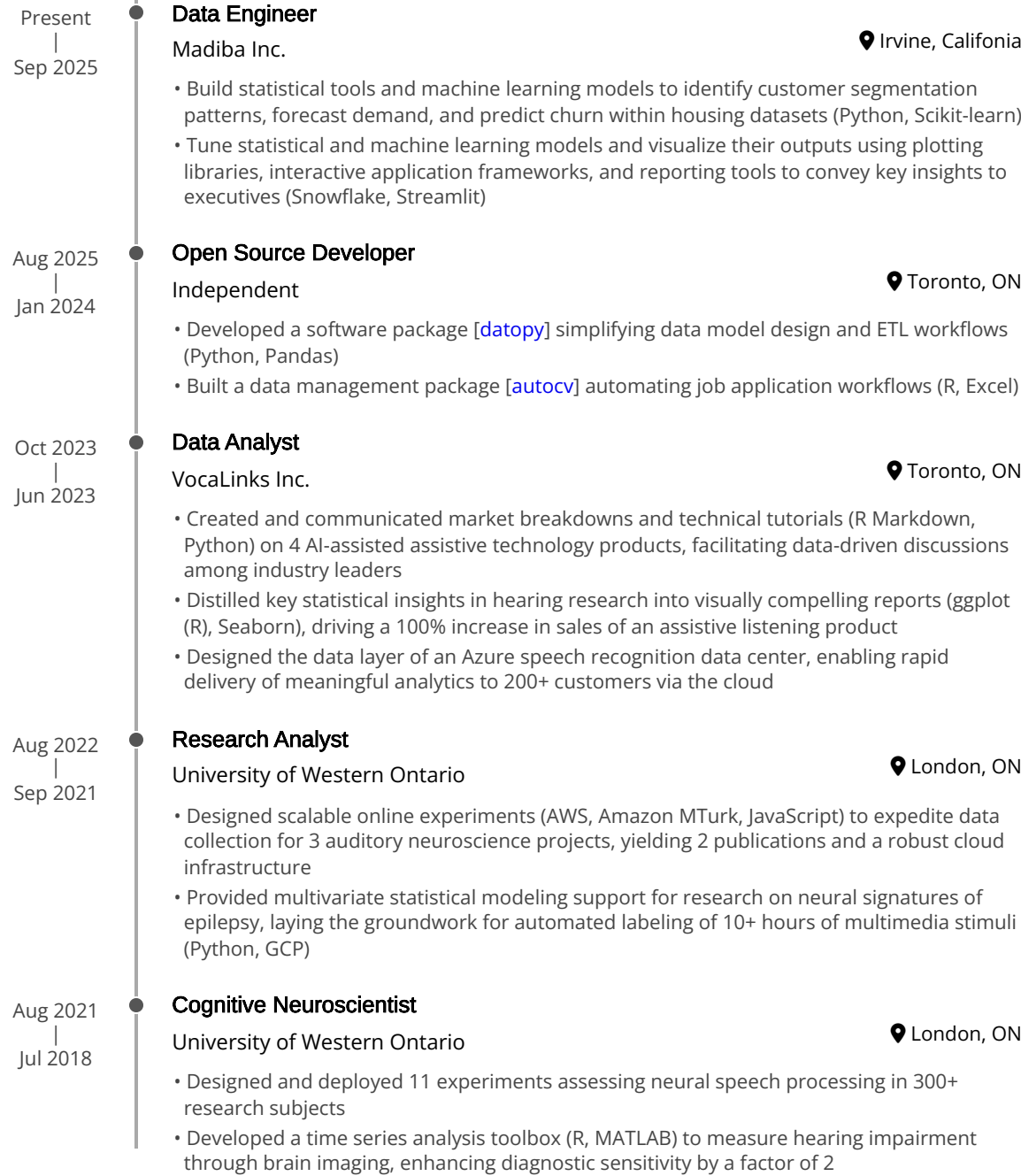
# Matthew Bain

## Data Engineer at Madiba Inc.

Data professional with 5+ years of experience building statistical solutions at scale. My work spans neuroscience, assistive tech, and housing, with a focus on statistically sound, interpretable machine learning. I help organizations understand their data and automate the hard stuff.



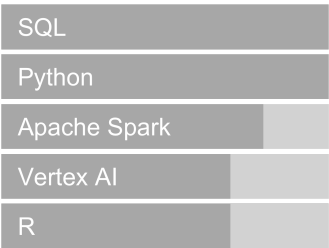
## Professional Experience



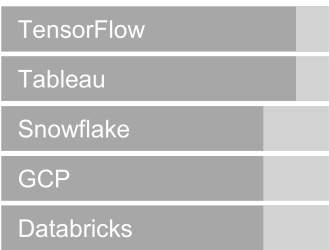
## Contact

Ontario  
 [Personal Website](#)  
 [GitHub](#)  
 [LinkedIn](#)

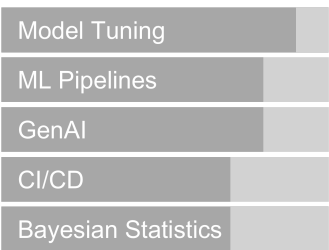
## Core tools



## Platforms



## Key competencies



Last updated on 2025-12-04.

Apr 2021  
|  
Sep 2018

● **Graduate Teaching Assistant**  
University of Western Ontario 📍 London, ON

- Directed 4 weekly research methods and statistics labs for 100+ undergraduate neuroscience students, leading to a nomination recognizing excellence in teaching and mentorship

## 🎓 Education

Dec 2023  
|  
Sep 2021

● **BSc (Honours), Mathematics and Statistics**  
McMaster University 📍 Hamilton, ON

Aug 2020  
|  
Sep 2018

● **MSc, Neuroscience** [[Thesis](#)]  
University of Western Ontario 📍 London, ON

Apr 2018  
|  
Sep 2014

● **BSc (Honours), Neuroscience**  
McMaster University 📍 Hamilton, ON

## 🏆 Certifications

Mar 2027  
|  
Mar 2025

● **Databricks Certified Data Engineer Associate** [[Certificate](#)]  
Databricks

Feb 2027  
|  
Feb 2025

● **Tableau Certified Data Analyst** [[Certificate](#)]  
Tableau

Expected  
Feb 2026

● **Professional Machine Learning Engineer**  
Google Cloud Platform (GCP)

Dec 2024

● **Machine Learning Specialization** [[Certificate](#)]  
Stanford University

## 🔗 Selected Projects

Present  
|  
Jan 2024

● **datopy (Data Management Python Package)** [[Documentation](#)]

- Implemented and maintain a package for handling unstructured data, providing a simple interface for data modeling, extraction, validation, and building ETL pipelines (Pandas, PyTest, Pydantic, GitHub Actions)

Present  
|  
Apr 2024

● **mlvizz (Machine Learning Data Visualization Package)**

- Build Python interfaces for efficient, intuitive model selection, tuning, inspection, and ML pipelines through modular, object-oriented designs with built-in data validation (Scikit-learn, TensorFlow, ArviZ, PyMC, SciPy)

Present  
|  
Apr 2024

● **statvizz (Statistical Data Visualization Package)**

- Build Python extensions unifying Pandas data summarization with Seaborn statistical plotting functionality for a unified, intuitive, fully transparent statistical plotting interface (Pandas, Seaborn, SciPy, Matplotlib)

Present  
|  
Jul 2024

● **mathvizz (Mathematical Visualization Package)**

- Build Python interfaces for exploring the geometry of machine learning math, including linear maps, derivatives, statistical distributions, and series approximations (SymPy, SciPy, Seaborn, Matplotlib)

Feb 2025

Feb 2025

Jul 2024

May 2024

●

One-stop Superstore Dashboard [\[Tableau\]](#)

- Designed an interactive dashboard displaying sales in a convenient, unified view, with key KPIs and sales logs visualized in both space and time (SQL, Excel)

●

autocv (Automated CV-generation R Package) [\[Documentation\]](#)

- Built a data management package automating job application workflows using validated data stores and macro-integrated spreadsheets, resulting in an efficient, extensible workflow (dplyr (R), pkgdown, Excel)